

Functions Part 2

Section 2

Chapter 4

An Introduction to Programming Using Python
David I. Schneider

Functions Calling Other Functions

- Function can call another function
- When the called function terminates
 - Control returns to the place in calling function just after where function call occurred.
- Example 1 : Function *firstPart* calls the function *secondPart*.

Functions Returning Multiple Values

- Functions can return any type of object
 - Function can return a tuple.
- Example 2: *balanceAndInterest* function returns a tuple
 - Giving values associated with deposit into savings account.

List Comprehension

- Simpler way to apply a certain function to each item of a list
 - Use list comprehension
- The *for* clause in a list comprehension can optionally be followed by an *if* clause.

Default Values

- Parameters of a function can have default values
 - Assigned to them when no values are passed to them
- Format for definition using default values

Passing by Parameter Name

- Arguments can be passed to functions by using names of the corresponding parameters
 - Instead of relying on position

Custom Sorting

- Functions can be used to order the items by any criteria we choose
- With a list of strings, we can sort them by
 - Length
 - Last characters
 - Number of vowels
 - ... by many other properties

Lambda Expressions

- One-line mini-functions
 - Can be used where a simple function is required.
 - Compute a single expression
 - Cannot be used as a replacement for complex functions
- Format
 - Where *expression* is the value to be returned

The *sorted* Function

- Contrast with *sort* function
 - *sort* function alters order of items in a list
 - *sorted* function returns a new ordered list
 - Both can make use of the optional arguments *key* and *reverse*.
 - *sorted* function also can be used with lists, strings, and tuples

End

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